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Abstract

A microbial-driven approach for effluent treatment, recycling, and management of Pharmaceutical and Personal Care Products (PPCPs) has been undertaken to mitigate the menace of water contamination. Bioremediation processes are mainly considered the first preference in pharmaceutical wastewater recycling and management. PPCPs are reported as one of the primary sources of emerging contaminants in various water matrices, which raises concern and requires efficient management. Their widespread utilization, persistently high level, and resistance to breaking down make them one of the potentially dangerous compounds causing harm to the ecosystem. Continually increasing PPCPs level PPCPs contaminants in water bodies raised concern for human health as they can produce potential risks with harmful and untoward impacts on our health. PPCPs are composed of multiple diverse compounds used by humans and animals, which include biopharmaceuticals, vitamins and nutritional supplements, antibiotics, counter-prescription drugs, cosmetics products, and unused pharmaceutical products. Personal care products are found to be bioaccumulative, reduce water quality and potentially impact ecological health. However, continual exposure to PPCPs in aquatic organisms, impacts their endocrine function disruption, gene toxicity, and antibiotic resistance. Decreased water quality may result in an outbreak of various water-borne diseases, which could have acute or long-term health complications and may result in an outbreak of various water-borne diseases, which could have acute or long-term effects on public and community health. Polluted water consumption by humans and animals produces serious health hazards and increased susceptibility to water-borne diseases such as carcinogenic organic or inorganic contaminants and infectious pathogens present in water bodies. Many water resource recovery facilities working on various conventional and advanced methods involve the utilization of microbes for filtration and advanced oxidation processes. Therefore, there is an immense need for bioremediation techniques facilitated by mixed cultures of bacteria, algae, and other microbes that can be used as an alternative approach for removing pharmaceutical content from effluent. This review highlights the various sources of PPCPs and their impacts on soil and water bodies, resulting in bioaccumulation. Different techniques are utilized to detect PPCPs, and various control strategies imply controlling, recycling, and managing waste.